Tarea3_Hermosilla_Moncada

July 5, 2025

- sexo: sexo del estudiante (1: mujer)
- edad: edad del estudiante (meses)
- imce: indice de masa corporal estandarizado (puntaje z)
- vive_padre: 1 si el padre vive en el hogar (2 indica que tiene mas de un padre)
- vive_madre: 2 si la madre vive en el hogar (2 indica que tiene mas de una madre)
- area: urbana=1, rural=0
- sk1: muestra afecto a padres (1: siempre - 5: nunca)
- sk2: muestra afecto a sus pares (1: siempre - 5: nunca)
- sk3: expresa sus sentimientos (1: siempre - 5: nunca)
- sk4: usa gestos para mostrar sentimientos (1: siempre - 5: nunca)
- sk5: juega con otros (1: siempre - 5: nunca)
- sk6: comparte sus cosas con otros (1: siempre - 5: nunca)
- sk7: es agresivo (1: siempre - 5: nunca)
- sk8: participa en juegos grupales (1: siempre - 5: nunca)
- sk9: hace preguntas a adultos (1: siempre - 5: nunca)
- sk10: tiene interes por libros (1: siempre - 5: nunca)
- sk11: tiene interes por su entorno (1: siempre - 5: nunca)
- sk12: juega a armar y desarmar cosas (1: siempre - 5: nunca)
- sk13: tiene expresiones artisticas (1: siempre - 5: nunca)
- act_fisica: frecuencia actividad fisica (1: nunca - 5: 5 o mas veces a la semana)
- educm: años de escolaridad de la madre
- educp: años de escolaridad del padre
- madre_work: si la madre trabaja (-1: labor domestica, 0: desempleada, 1: empleada)

1.

```
               sexo          edad          imce    vive_padre    vive_madre  \
count  31055.000000  31055.000000  31055.000000  31055.000000  31055.000000
mean       0.538979     82.589084      1.017787      0.720335      0.966511
std        0.498486      4.360270      1.370552      0.449415      0.189671
min        0.000000     62.000000     -5.020000      0.000000      0.000000
25%        0.000000     80.000000      0.120000      0.000000      1.000000
50%        1.000000     81.000000      0.970000      1.000000      1.000000
75%        1.000000     82.000000      1.930000      1.000000      1.000000
max        1.000000    107.000000      5.040000      2.000000      2.000000

                sk1           sk2           sk3           sk4           sk5  \
count  31055.000000  31055.000000  31055.000000  31055.000000  31055.000000
mean       1.114088      1.391177      1.262856      1.256352      1.268588
```

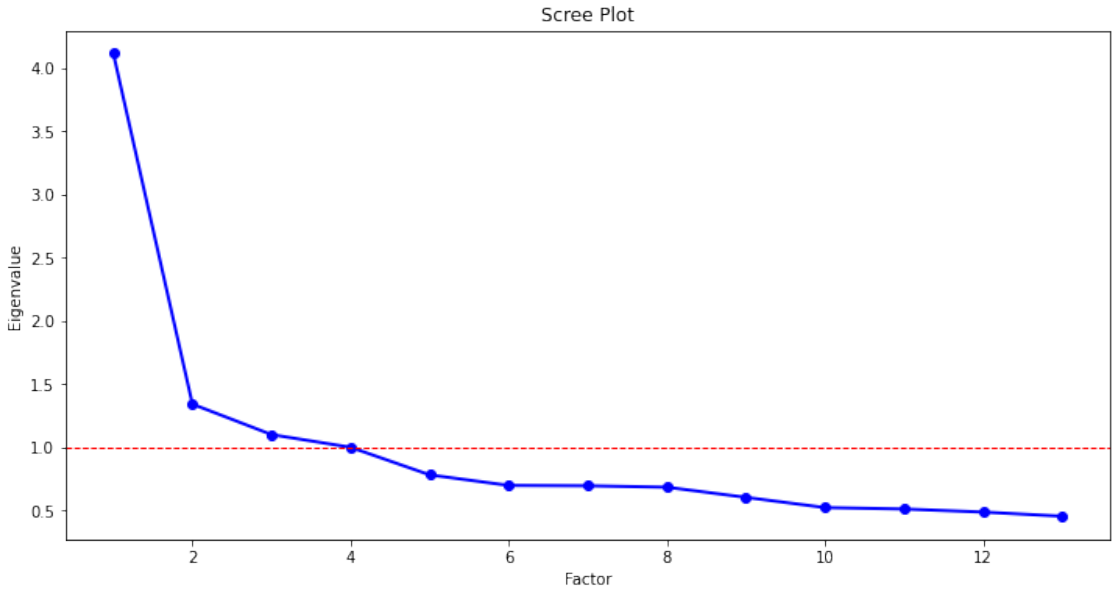
```
std        0.394004      0.650190      0.585766      0.578266      0.564929
min        1.000000      1.000000      1.000000      1.000000      1.000000
25%        1.000000      1.000000      1.000000      1.000000      1.000000
50%        1.000000      1.000000      1.000000      1.000000      1.000000
75%        1.000000      2.000000      1.000000      1.000000      1.000000
max        5.000000      5.000000      5.000000      5.000000      5.000000

       …           sk9          sk10          sk11          sk12  \
count  …  31055.000000  31055.000000  31055.000000  31055.000000
mean   …      1.333666      1.853743      1.385349      1.502785
std    …      0.666009      0.941171      0.668783      0.799375
min    …      1.000000      1.000000      1.000000      1.000000
25%    …      1.000000      1.000000      1.000000      1.000000
50%    …      1.000000      2.000000      1.000000      1.000000
75%    …      1.000000      2.000000      2.000000      2.000000
max    …      5.000000      5.000000      5.000000      5.000000

               sk13    act_fisica          area         educm         educp  \
count  31055.000000  30051.000000  31055.000000  30682.000000  31055.000000
mean       1.694735      2.556620      0.911544     13.069291     12.955756
std        0.984946      1.068041      0.283961      3.338320      3.425190
min        1.000000      1.000000      0.000000      0.000000      0.000000
25%        1.000000      2.000000      1.000000     12.000000     11.000000
50%        1.000000      2.000000      1.000000     13.000000     13.000000
75%        2.000000      3.000000      1.000000     15.000000     14.000000
max        5.000000      5.000000      1.000000     22.000000     22.000000

         madre_work
count  31055.000000
mean       0.092803
std        0.943509
min       -1.000000
25%       -1.000000
50%        0.000000
75%        1.000000
max        1.000000

 [8 rows x 23 columns]

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 31055 entries, 0 to 31054
Data columns (total 13 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   sk1     31055 non-null  int64
 1   sk2     31055 non-null  int64
 2   sk3     31055 non-null  int64
 3   sk4     31055 non-null  int64
```

```
 4   sk5     31055 non-null  int64
 5   sk6     31055 non-null  int64
 6   sk7     31055 non-null  int64
 7   sk8     31055 non-null  int64
 8   sk9     31055 non-null  int64
 9   sk10    31055 non-null  int64
 10  sk11    31055 non-null  int64
 11  sk12    31055 non-null  int64
 12  sk13    31055 non-null  int64
dtypes: int64(13)
memory usage: 3.1 MB
```

No existen null en los sk's, por lo que un *dropna* no será necesario, además se invierte el sentido de sk7 ya que es de connotación negtiva, más adelante veremos que variables son significativas para cada modelo y cual podemos descartar.

2.

```
array([[0.183970, 0.549198, 0.155598],
       [0.177203, 0.517433, 0.350410],
       [0.182944, 0.596588, 0.168045],
       [0.198779, 0.686060, 0.200987],
       [0.121309, 0.219980, 0.724979],
       [0.173086, 0.210702, 0.481159],
       [-0.069524, -0.074626, -0.136478],
       [0.269342, 0.104813, 0.478344],
       [0.495931, 0.215337, 0.207927],
       [0.563365, 0.122575, 0.127228],
       [0.664883, 0.209988, 0.205834],
       [0.526984, 0.116246, 0.144172],
       [0.485171, 0.154101, 0.134872]])

array([4.114952, 1.341624, 1.100575, 1.000161, 0.781977, 0.698537,
       0.695787, 0.684453, 0.604042, 0.523209, 0.512081, 0.487796,
       0.454807])
```



Vemos claramente como se forma un codo con dos factores, por lo que tomaremos esta configuración como primera opción:

```
array([[0.544433, 0.168621],
       [0.639659, 0.193829],
       [0.578932, 0.175454],
       [0.661521, 0.195791],
       [0.499339, 0.254957],
       [0.417627, 0.245437],
       [-0.137482, -0.085336],
       [0.323907, 0.338242],
       [0.273662, 0.511502],
       [0.151393, 0.563508],
       [0.262579, 0.673274],
       [0.152455, 0.537187],
       [0.183986, 0.488821]])

array([0.173059, 0.322447])

array([0.324840, 0.446733, 0.365946, 0.475945, 0.314343, 0.234651,
       0.026184, 0.219323, 0.336525, 0.340461, 0.522246, 0.311813,
       0.272797])
```

Vemos en [9] que la varianza acumulada es de un 32%, lo que indica que con dos factores no estamos explicando una gran parte de la varianza de los datos. Por lo que se opta por el criterio de dejar todos aquellos factores con un valor propio mayor a 1

```
1.0001606732123345
```

Si bien esta muy al límite alcanza a superar uno por lo que en este modelo usaremos 4 factores.

```
array([[0.179220, 0.521750, 0.062918, 0.315360],
       [0.173856, 0.486274, 0.258577, 0.369580],
       [0.199616, 0.632740, 0.171230, -0.029259],
       [0.216855, 0.695906, 0.191845, 0.039809],
       [0.138459, 0.228331, 0.740685, 0.128042],
       [0.176714, 0.178838, 0.412680, 0.313400],
       [-0.058745, -0.029790, -0.062723, -0.295514],
       [0.287356, 0.116554, 0.464761, 0.067503],
       [0.507360, 0.214436, 0.190703, 0.045422],
       [0.565324, 0.094034, 0.071719, 0.169465],
       [0.667603, 0.194712, 0.165143, 0.123248],
       [0.533830, 0.115109, 0.130746, 0.026176],
       [0.487336, 0.142442, 0.105435, 0.087389]])
```

```
sk1: 0.4078
sk2: 0.4701
sk3: 0.4704
sk4: 0.5697
sk5: 0.6363
sk6: 0.3317
sk7: 0.0956
sk8: 0.3167
sk9: 0.3418
sk10: 0.3623
sk11: 0.5261
sk12: 0.3160
sk13: 0.2765
```

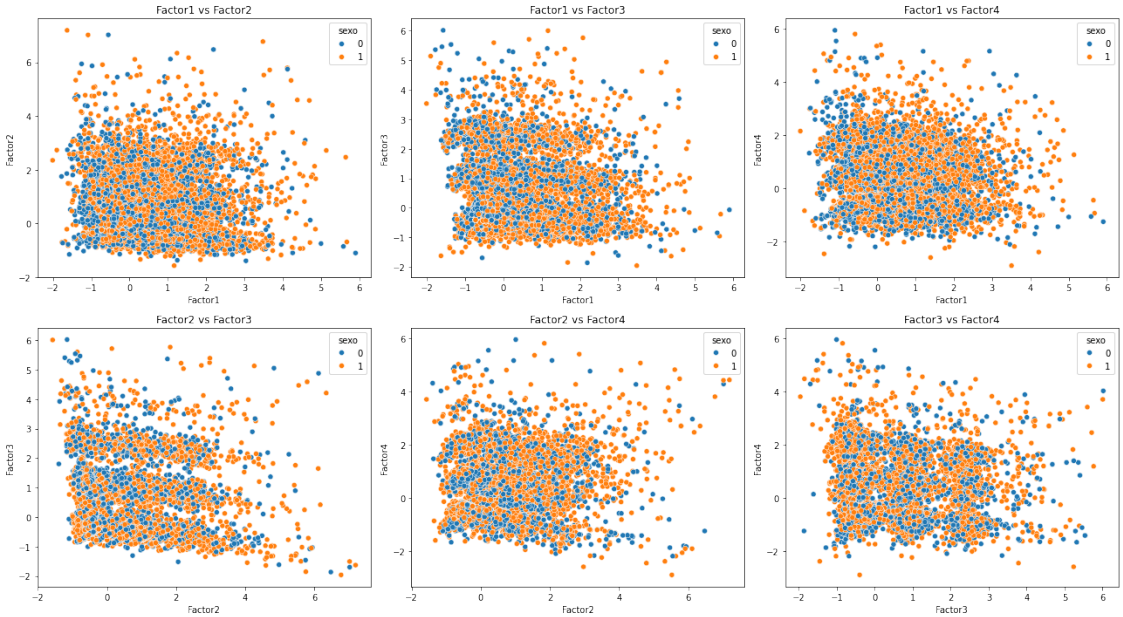
Vemos acá que con 4 factores (lo cual es harto) se explica tan solo un 9% de la varianza de sk7 por lo que lo podemos conciderar descartable, ya que no entra en ningun grupo.

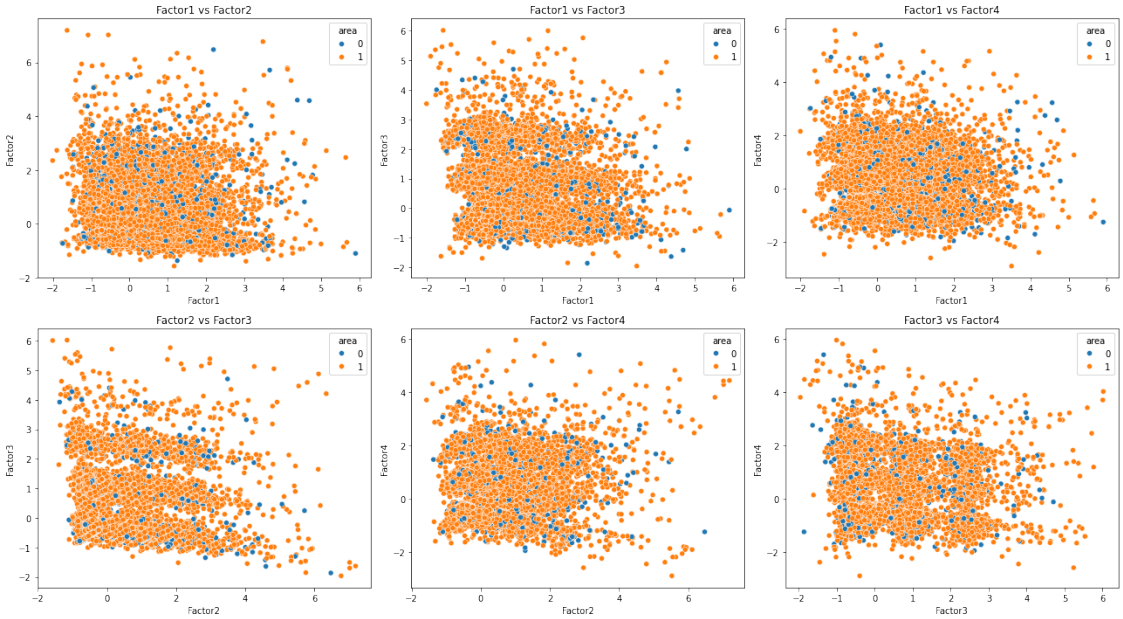
```
array([[0.199214, 0.428295, 0.114913, 0.333055],
       [0.167053, 0.329436, 0.277316, 0.727069],
       [0.187375, 0.597364, 0.170896, 0.130160],
       [0.196316, 0.735401, 0.194389, 0.133002],
       [0.124426, 0.194595, 0.790709, 0.105628],
       [0.197004, 0.164895, 0.421211, 0.176455],
       [0.283984, 0.099812, 0.455804, 0.074725],
       [0.503883, 0.229603, 0.199611, 0.026651],
       [0.571486, 0.079935, 0.096713, 0.118272],
       [0.668488, 0.191173, 0.184205, 0.093979],
       [0.528177, 0.110625, 0.135921, 0.043881],
       [0.488924, 0.122475, 0.115983, 0.102992]])

(array([1.821570, 1.387534, 1.282555, 0.758128]),
 array([0.151798, 0.115628, 0.106880, 0.063177]),
```

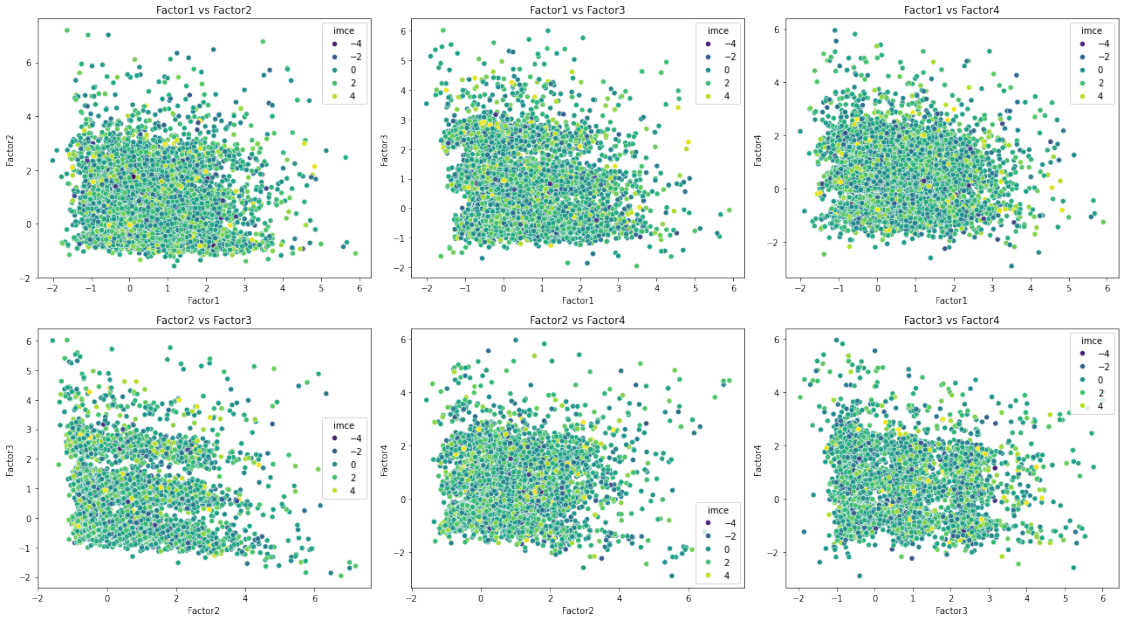
```
array([0.151798, 0.267425, 0.374305, 0.437482]))
```

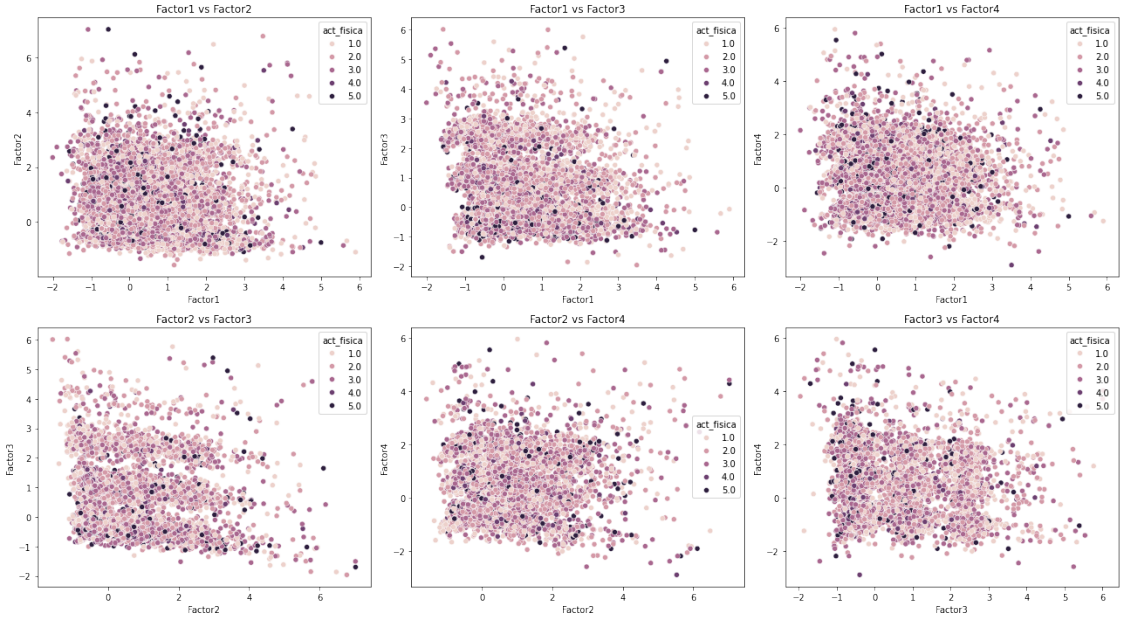
Ahora con 4 podemos explicar un 44%, mejora pero no es tan bueno aún, aunque lo dejaremos así ya que aumentar tantos factores no es una buena práctica y tampoco aumentaría tanto este valor.

3.









Vemos que ningun gráfico presenta segrecación de los factores, vemos solo ruido.

4.

```
array([[0.199214, 0.428295, 0.114913, 0.333055],
       [0.167053, 0.329436, 0.277316, 0.727069],
       [0.187375, 0.597364, 0.170896, 0.130160],
```

```
        [0.196316, 0.735401, 0.194389, 0.133002],
        [0.124426, 0.194595, 0.790709, 0.105628],
        [0.197004, 0.164895, 0.421211, 0.176455],
        [0.283984, 0.099812, 0.455804, 0.074725],
        [0.503883, 0.229603, 0.199611, 0.026651],
        [0.571486, 0.079935, 0.096713, 0.118272],
        [0.668488, 0.191173, 0.184205, 0.093979],
        [0.528177, 0.110625, 0.135921, 0.043881],
        [0.488924, 0.122475, 0.115983, 0.102992]])
```

Como elegimos entre 3 y 4 factores tan al límite, vamos a ver que modelo es mejor. El de tres factores se basa en los loadings que vimos por defecto en una primera instancia que venia de a 3 factores, y el de 4, es con el que nos quedamos en la pregunta 2

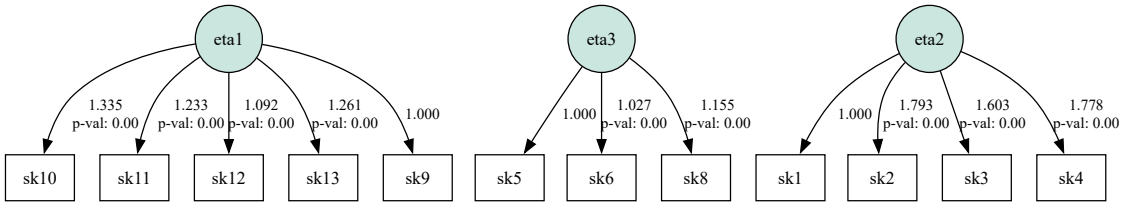
```
       DoF  DoF Baseline         chi2  chi2 p-value  chi2 Baseline       CFI  \
Value   51            66  4401.365973           0.0   87611.052581  0.950307

            GFI      AGFI       NFI       TLI     RMSEA        AIC  \
Value  0.949762  0.934987  0.949762  0.935692  0.052411  53.716544

              BIC    LogLik
Value  278.991452  0.141728
-----
       DoF  DoF Baseline         chi2  chi2 p-value  chi2 Baseline       CFI  \
Value   48            66  3567.641916           0.0   87611.052581  0.959796

            GFI      AGFI       NFI      TLI     RMSEA        AIC        BIC  \
Value  0.959279  0.944008  0.959279  0.94472  0.048593  59.770237  310.07569

         LogLik
Value  0.114881
```

Ambos tienen un chi2 p-value de 0, por lo que son significativos, pero el modelo 1 tiene un menor bic por lo que nos aseguramos que es un modelo sencillo y sin sobreajuste, aunque bien este beneficio es muy marginal al del modelo 2

```
Name of objective: MLW
Optimization method: SLSQP
Optimization successful.
Optimization terminated successfully
Objective value: 0.115
Number of iterations: 40
Params: 1.330 1.229 1.089 1.257 1.649 1.841 1.043 1.161 0.161 0.055 0.096 0.104
0.055 0.092 0.163 0.056 0.139 0.143 0.101 0.601 0.204 0.448 0.716 0.279 0.194
0.149 0.156 0.373 0.477 0.283
```

	lval	op	rval	Estimate	Est. Std	Std. Err	z-value	p-value
0	sk9	~	eta1	1.000000	0.600501	-	-	-
1	sk10	~	eta1	1.334713	0.567130	0.017645	75.641962	0.0

2	sk11	~	eta1	1.232807	0.737146	0.013993	88.102986	0.0
3	sk12	~	eta1	1.091950	0.546278	0.014832	73.622089	0.0
4	sk13	~	eta1	1.261417	0.512138	0.017981	70.151539	0.0
5	sk1	~	eta2	1.000000	0.595909	-	-	-
6	sk2	~	eta2	1.793157	0.647690	0.021586	83.072223	0.0
7	sk3	~	eta2	1.603422	0.642710	0.019395	82.673817	0.0
8	sk4	~	eta2	1.777960	0.721890	0.020174	88.132298	0.0
9	sk5	~	eta3	1.000000	0.719657	-	-	-
10	sk6	~	eta3	1.026632	0.562749	0.013824	74.266085	0.0
11	sk8	~	eta3	1.155310	0.562710	0.015557	74.262593	0.0
12	eta1	~~	eta1	0.159913	1.000000	0.003116	51.314941	0.0
13	eta1	~~	eta2	0.056757	0.604481	0.001019	55.701576	0.0
14	eta1	~~	eta3	0.096470	0.593335	0.001696	56.871328	0.0
15	eta2	~~	eta2	0.055129	1.000000	0.001076	51.236155	0.0
16	eta3	~~	eta3	0.165310	1.000000	0.002849	58.023053	0.0
17	eta3	~~	eta2	0.062711	0.656901	0.00104	60.278918	0.0
18	sk1	~~	sk1	0.100118	0.644893	0.000947	105.731758	0.0
19	sk10	~~	sk10	0.600837	0.678363	0.005587	107.548234	0.0
20	sk11	~~	sk11	0.204229	0.456616	0.002503	81.593079	0.0
21	sk12	~~	sk12	0.448268	0.701580	0.0041	109.342606	0.0
22	sk13	~~	sk13	0.715675	0.737715	0.006396	111.894751	0.0
23	sk2	~~	sk2	0.245293	0.580497	0.002461	99.659864	0.0
24	sk3	~~	sk3	0.201387	0.586924	0.002007	100.32833	0.0
25	sk4	~~	sk4	0.160143	0.478875	0.001843	86.908808	0.0
26	sk5	~~	sk5	0.153880	0.482095	0.002146	71.703273	0.0
27	sk6	~~	sk6	0.375940	0.683313	0.003685	102.011723	0.0
28	sk8	~~	sk8	0.476185	0.683358	0.004668	102.016687	0.0
29	sk9	~~	sk9	0.283549	0.639398	0.002721	104.22048	0.0

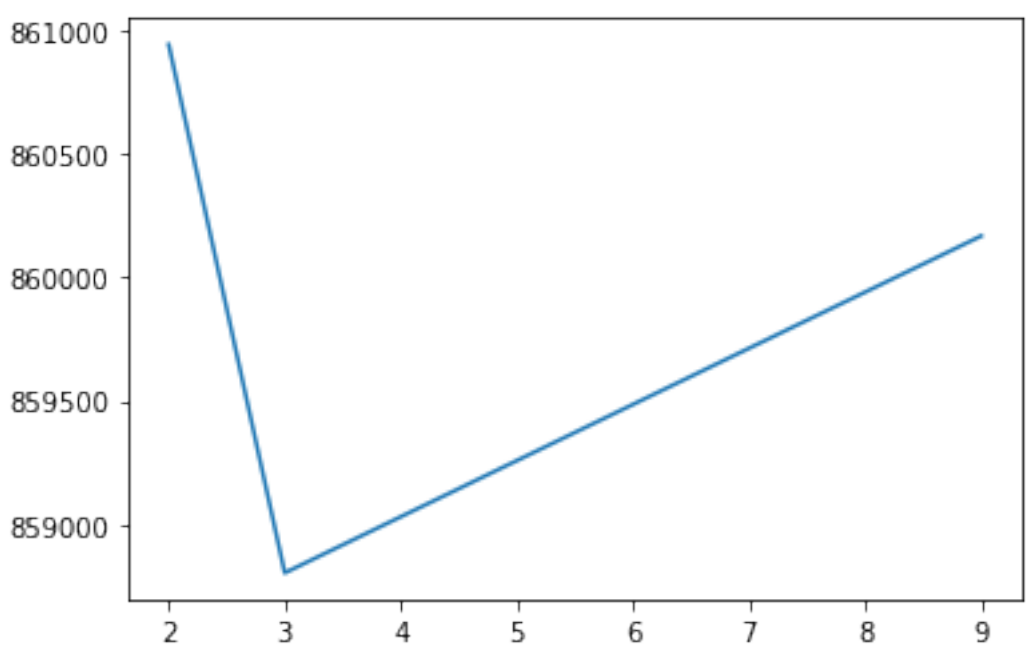


La magnitud de los parametros del modelo indican relaciones saludables entre las variables latentes y las observadas, además ninguna presenta un p-value>0.5

5.

```
Fitting StepMix…

Initializations (n_init) : 100%|      | 1/1 [00:02<00:00,  2.62s/it,
max_LL=-4.3e+5, max_avg_LL=-13.9]

Fitting StepMix…
```

```
Initializations (n_init) : 100%|      | 1/1 [00:05<00:00,  5.81s/it,
max_LL=-4.29e+5, max_avg_LL=-13.8]

Fitting StepMix…

Initializations (n_init) : 100%|      | 1/1 [00:12<00:00, 12.78s/it,
max_LL=-4.29e+5, max_avg_LL=-13.8]

Fitting StepMix…

Initializations (n_init) : 100%|      | 1/1 [00:13<00:00, 13.62s/it,
max_LL=-4.29e+5, max_avg_LL=-13.8]

Fitting StepMix…

Initializations (n_init) : 100%|      | 1/1 [00:10<00:00, 10.66s/it,
max_LL=-4.29e+5, max_avg_LL=-13.8]

Fitting StepMix…

Initializations (n_init) : 100%|      | 1/1 [00:11<00:00, 11.51s/it,
max_LL=-4.29e+5, max_avg_LL=-13.8]

Fitting StepMix…

Initializations (n_init) : 100%|      | 1/1 [00:14<00:00, 14.31s/it,
max_LL=-4.29e+5, max_avg_LL=-13.8]

Fitting StepMix…

Initializations (n_init) : 100%|      | 1/1 [00:09<00:00,  9.84s/it,
max_LL=-4.29e+5, max_avg_LL=-13.8]

<Axes: >
```

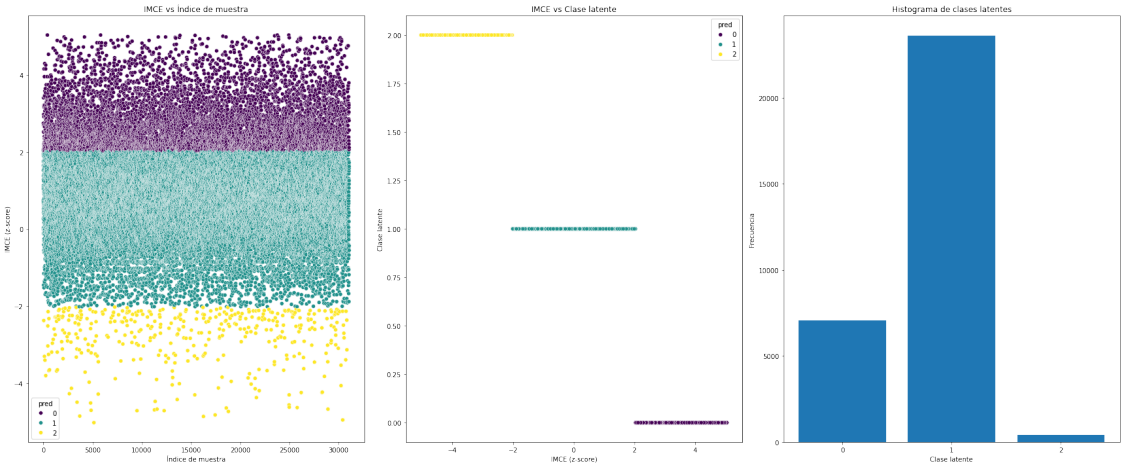


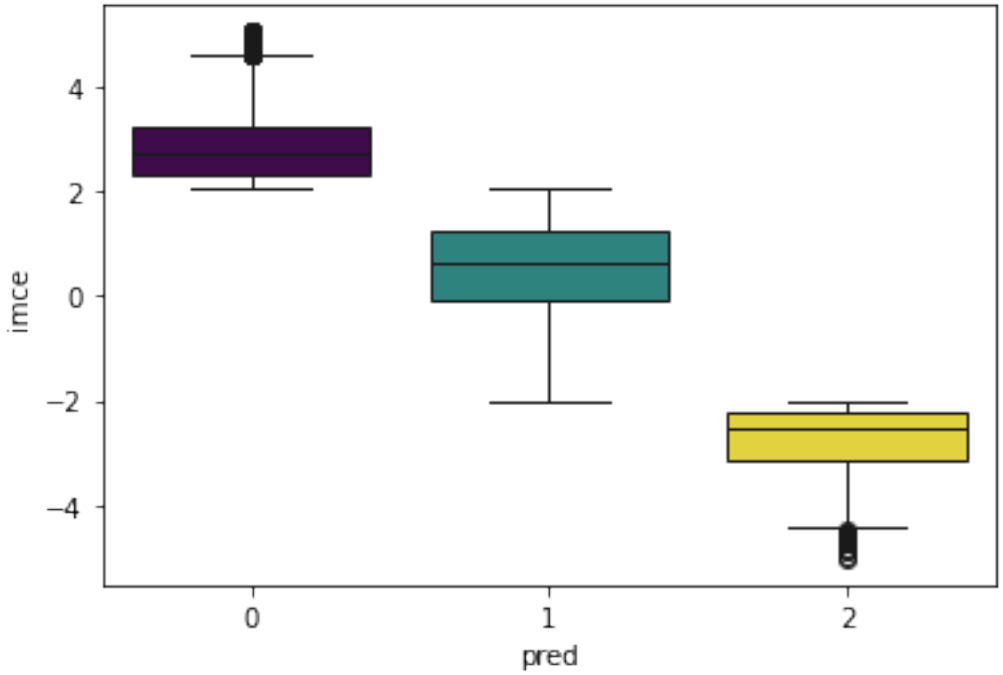
En este set de datos ocurre algo muy curioso, ya que despues de 3 componentes el BIC no sigue disminuyendo si no que aumenta, por lo que simplemente nos quedaremos con 3.

```
Fitting StepMix…

Initializations (n_init) : 100%|      | 1/1 [00:02<00:00,  2.35s/it,
max_LL=-5.37e+4, max_avg_LL=-1.73]

================================================================================
MODEL REPORT
================================================================================
    ============================================================================
    Measurement model parameters
    ============================================================================
          model_name     gaussian_unit
          class_no                   0       1       2
          param variable
          means imce            2.4074  0.5922 -1.9669
    ============================================================================
    Class weights
    ============================================================================
        Class 1 : 0.27
        Class 2 : 0.71
        Class 3 : 0.02
    ============================================================================
    Fit for 3 latent classes
    ============================================================================
    Estimation method             : 1-step
    Number of observations        : 31055
    Number of latent classes      : 3
    Number of estimated parameters: 5
    Log-likelihood (LL)           : -53735.8888
    -2LL                          : 107471.7775
    Average LL                    : -1.7303
    AIC                           : 107481.78
    BIC                           : 107523.50
    CAIC                          : 107528.50
    Sample-Size Adjusted BIC      : 107559.32
    Entropy                       : 11958.6120
    Scaled Relative Entropy       : 0.6495


/var/folders/1b/_0my04wn0l5fq3m24vckmcmc0000gn/T/ipykernel_50282/732862712.py:5:
SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead
```

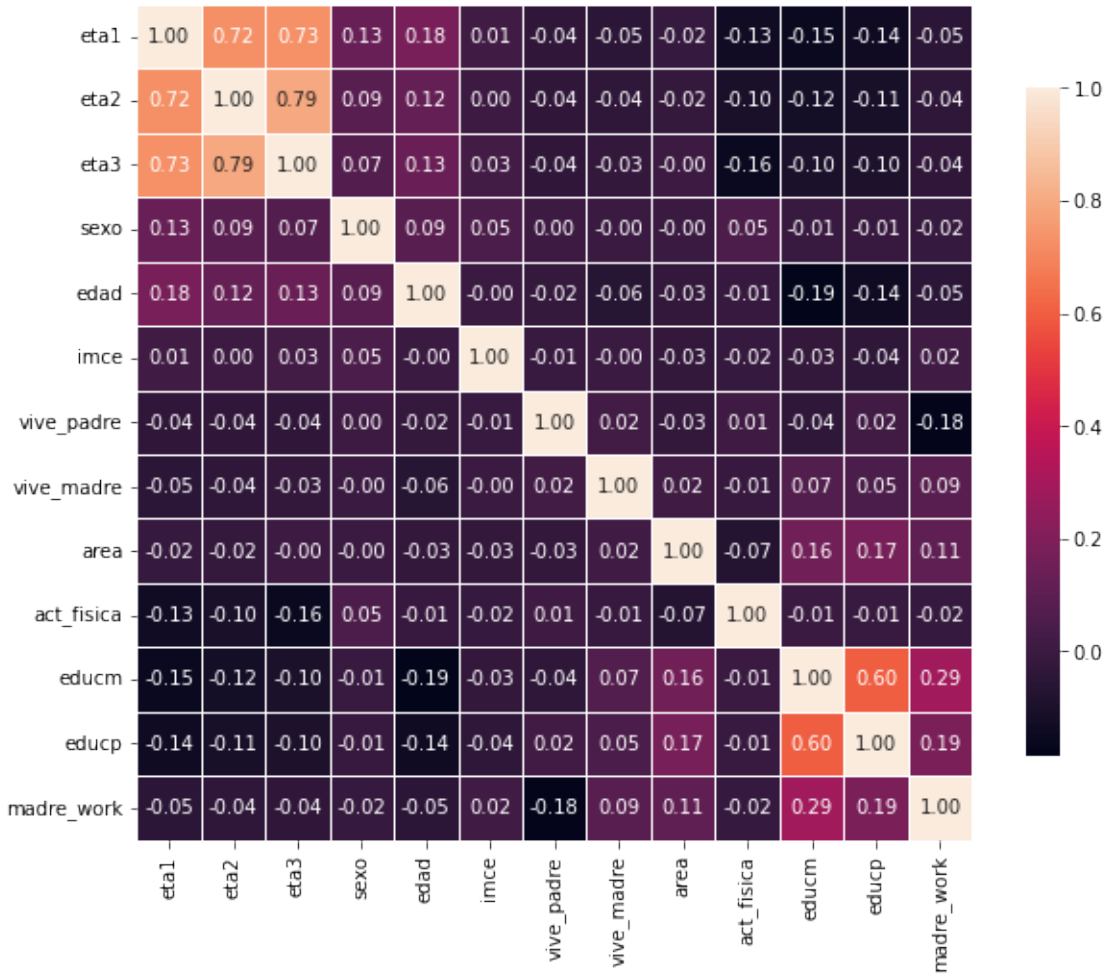
```
See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  df_cl['pred']= model.predict(df_cl)
```

La magnitud de la Scaled Relative Entropy nos indica que el modelo es bueno y vemos que como el imc está normalizado la clase 2 va hasta -2, y la clase 0 desde 2.4, estando la gran mayoria de los datos en la clase 1 como se ve en los pesos, existe además mucho más sobrepeso que desnutrición.

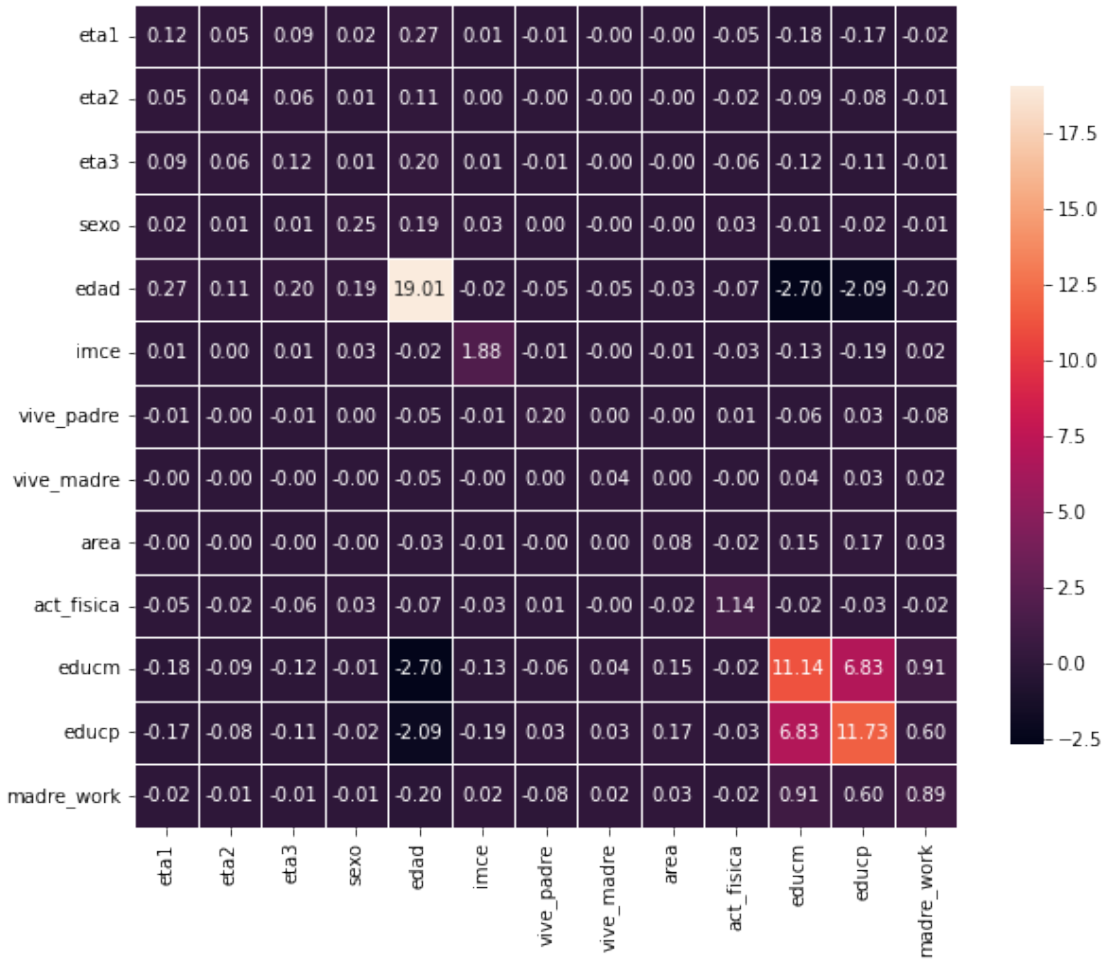




6.




```
          eta1      eta2      eta3      sexo      edad      imce  vive_padre  \
eta1  0.124746  0.053329  0.089602  0.022719  0.271444  0.006717   -0.005608
eta2  0.053329  0.043456  0.057464  0.009390  0.112088  0.001403   -0.003577
eta3  0.089602  0.057464  0.120488  0.012452  0.203619  0.011996   -0.005960

      vive_madre      area  act_fisica     educm     educp  madre_work
eta1   -0.003487 -0.001796   -0.049381 -0.181549 -0.166691   -0.017200
eta2   -0.001604 -0.001266   -0.022780 -0.085819 -0.078943   -0.008274
eta3   -0.001896 -0.000033   -0.056984 -0.119450 -0.113635   -0.011498
```



Se identifica del modelo anterior, que: - eta1 : desarrollo sentimental - eta2 : desarrollo lúdico y social - eta3 : desarrollo intelectual

Por lo que en base a esa interpretación y respaldado por la matriz de correlación, se propone el siguiente modelo:

```
Name of objective: MLW
Optimization method: SLSQP
Optimization successful.
Optimization terminated successfully
Objective value: 0.909
Number of iterations: 102
Params: 1.372 1.208 1.073 1.280 2.582 -0.289 -1.861 -1.751 1.785 1.599 1.769
1.030 1.185 -0.595 0.179 0.636 1.060 0.207 -0.063 -0.037 -0.027 -0.219 1.101
0.081 18.080 10.600 11.241 0.068 0.019 0.060 0.091 0.886 0.245 0.100 0.587 0.215
0.456 0.710 0.246 0.201 0.161 0.157 0.378 0.469 0.285 0.036 0.202
```

```
       DoF  DoF Baseline          chi2  chi2 p-value  chi2 Baseline       CFI  \
Value  184           210  28214.166883           0.0  115521.968845  0.756919

            GFI      AGFI       NFI      TLI    RMSEA        AIC         BIC  \
Value  0.755768  0.721257  0.755768  0.72257  0.07004  92.182955  484.328165

         LogLik
Value  0.908523
```

EL CFI y el TLI debieran estar entre 0.90-0.95, para ser aceptable y RMSEA menor a 0.06 ya que este ultimo estima la discrepancia promedio por grado de libertad entre la matriz de covarianzas observada y la predicha por el modelo. En este caso no se cumple niingun criterio, por lo que obtenemos los parametros del modelo y vemos que es lo que esta relacionado y no es significante.

	lval	op	rval	Estimate	Est. Std	Std. Err	z-value	\
0	eta1	~	imce	1.000000	0.755971	-	-	
1	eta2	~	imce	0.635524	0.812799	0.011174	56.873146	
2	eta3	~	imce	1.059892	0.792396	0.01738	60.984253	
3	sk9	~	eta1	1.000000	0.598492	-	-	
4	sk10	~	eta1	1.371806	0.581000	0.01778	77.154548	
5	sk11	~	eta1	1.208448	0.720266	0.013767	87.778813	
6	sk12	~	eta1	1.073269	0.535196	0.014761	72.708068	
7	sk13	~	eta1	1.280246	0.518142	0.018042	70.960123	
8	edad	~	eta1	2.582246	0.236078	0.151134	17.085763	
9	edad	~	imce	-0.288548	-0.019943	0.206672	-1.396167	
10	educm	~	eta1	-1.861380	-0.222208	0.055095	-33.785193	
11	educp	~	eta1	-1.750918	-0.203807	0.056275	-31.113435	
12	sk1	~	eta2	1.000000	0.598033	-	-	
13	sk2	~	eta2	1.785127	0.646894	0.021448	83.230506	
14	sk3	~	eta2	1.599070	0.643201	0.019282	82.932612	
15	sk4	~	eta2	1.769055	0.720803	0.020027	88.335257	
16	sk5	~	eta3	1.000000	0.713486	-	-	
17	sk6	~	eta3	1.029801	0.559572	0.013897	74.0998	
18	sk8	~	eta3	1.184821	0.572050	0.015757	75.19194	
19	act_fisica	~	eta3	-0.595010	-0.224570	0.045649	-13.034426	
20	act_fisica	~	imce	0.178856	0.050468	0.060981	2.932998	
21	sexo	~	imce	0.206809	0.125021	0.011332	18.250353	
22	vive_padre	~	imce	-0.063460	-0.042552	0.010041	-6.320126	
23	vive_madre	~	imce	-0.036673	-0.058266	0.004246	-8.636736	
24	area	~	imce	-0.027216	-0.028884	0.006336	-4.295315	
25	madre_work	~	imce	-0.218538	-0.069800	0.021162	-10.327071	
26	eta1	~~	eta1	0.068091	0.428508	0.00184	36.996565	
27	eta2	~~	eta2	0.018841	0.339358	0.000615	30.654361	
28	eta3	~~	eta3	0.060457	0.372109	0.001962	30.814042	
29	imce	~~	imce	0.090811	1.000000	0.002307	39.355404	
30	act_fisica	~~	act_fisica	1.100630	0.964983	0.009082	121.194457	
31	area	~~	area	0.080564	0.999166	0.000647	124.567638	

32	edad	~~	edad	18.079609	0.950988	0.14796	122.192567
33	educm	~~	educm	10.599582	0.950624	0.086299	122.824493
34	educp	~~	educp	11.240768	0.958463	0.091299	123.12061
35	madre_work	~~	madre_work	0.885842	0.995128	0.007123	124.364461
36	sexo	~~	sexo	0.244606	0.984370	0.001976	123.814077
37	sk1	~~	sk1	0.099715	0.642356	0.000945	105.516348
38	sk10	~~	sk10	0.586823	0.662439	0.005496	106.781672
39	sk11	~~	sk11	0.215249	0.481217	0.002483	86.699244
40	sk12	~~	sk12	0.455990	0.713565	0.004122	110.625953
41	sk13	~~	sk13	0.709657	0.731529	0.006345	111.84239
42	sk2	~~	sk2	0.245855	0.581528	0.002464	99.769081
43	sk3	~~	sk3	0.201184	0.586292	0.002007	100.264168
44	sk4	~~	sk4	0.160668	0.480443	0.001844	87.14468
45	sk5	~~	sk5	0.156687	0.490937	0.002129	73.603797
46	sk6	~~	sk6	0.377967	0.686880	0.003687	102.515142
47	sk8	~~	sk8	0.468894	0.672759	0.004645	100.952364
48	sk9	~~	sk9	0.284720	0.641808	0.00271	105.047433
49	vive_madre	~~	vive_madre	0.035852	0.996605	0.000288	124.438999
50	vive_padre	~~	vive_padre	0.201606	0.998189	0.001619	124.518674

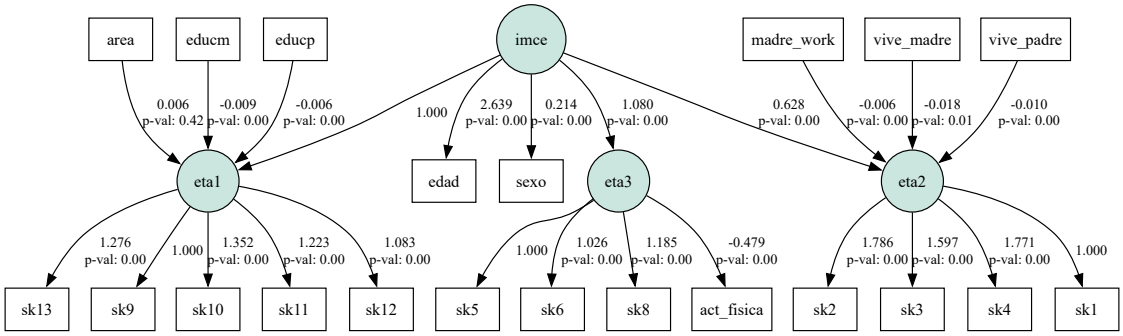
	p-value
0	-
1	0.0
2	0.0
3	-
4	0.0
5	0.0
6	0.0
7	0.0
8	0.0
9	0.162664
10	0.0
11	0.0
12	-
13	0.0
14	0.0
15	0.0
16	-
17	0.0
18	0.0
19	0.0
20	0.003357
21	0.0
22	0.0
23	0.0
24	0.000017
25	0.0

```
26       0.0
27       0.0
28       0.0
29       0.0
30       0.0
31       0.0
32       0.0
33       0.0
34       0.0
35       0.0
36       0.0
37       0.0
38       0.0
39       0.0
40       0.0
41       0.0
42       0.0
43       0.0
44       0.0
45       0.0
46       0.0
47       0.0
48       0.0
49       0.0
50       0.0
```

Ajustamos aquellas relaciones debiles, siempre teniendo en cuenta el contexto que ya dimos de cada eta y la matrices de correlaciones y covarinzas.

```
       DoF  DoF Baseline         chi2  chi2 p-value  chi2 Baseline       CFI  \
Value  192           216  10909.83632           0.0  115521.964821  0.907049

            GFI      AGFI       NFI      TLI     RMSEA        AIC         BIC  \
Value  0.905561  0.893756  0.905561  0.89543  0.042398  77.297386  402.694475

         LogLik
Value  0.351307
```

Se cumplen los criterios y el modelo queda así:


area
educm
educp
imce
madre_work
vive_madre
vive_padre
0.006
p-val: 0.42
-0.009
p-val: 0.00
-0.006
p-val: 0.00
1.000
2.639
p-val: 0.00
0.214
p-val: 0.00
1.080
p-val: 0.00
0.628
p-val: 0.00
-0.006
p-val: 0.00
-0.018
p-val: 0.01
-0.010
p-val: 0.00
eta1
edad
sexo
eta3
eta2
1.276
p-val: 0.00
1.000
1.352
p-val: 0.00
1.223
p-val: 0.00
1.083
p-val: 0.00
1.000
1.026
p-val: 0.00
1.185
p-val: 0.00
-0.479
p-val: 0.00
1.786
p-val: 0.00
1.597
p-val: 0.00
1.771
p-val: 0.00
1.000
sk13
sk9
sk10
sk11
sk12
sk5
sk6
sk8
act_fisica
sk2
sk3
sk4
sk1